In recent decades, interest in hypothetical time machine designs has revived. In the 1970s, Frank Tipler reanalyzed and refined van Stockum's solution, and in 1991, Richard Gott of Princeton University discovered another method for building a time machine making use of so-called cosmic strings (hypothetical, infinitely long, filamentary remnants of phase transitions in the early universe). These are all important contributions, but the proposal that's simplest to describe, using concepts we've developed in previous chapters, was found by Kip Thorne and his students at the California Institute of Technology. It makes use of wormholes.

Blueprint for a Wormhole Time Machine

I'll first lay out the basic strategy for constructing Thorne's wormhole time machine, and in the next section I'll discuss the challenges faced by any contractor Thorne might hire to execute the plans.

A *wormhole* is a hypothetical tunnel through space. A more familiar kind of tunnel, such as one that's been bored through the side of a mountain, provides a shortcut from one location to another. Wormholes serve a similar function, but they differ from conventional tunnels in one important respect. Whereas conventional tunnels provide a new route through existing space—the mountain and the space it occupies exist before a tunnel is constructed—a wormhole provides a tunnel from one point in space to another along a new, previously nonexistent tube of space. Were you to remove the tunnel through the mountain, the space it occupied would still exist. Were you to remove a wormhole, the space it occupied would vanish.

Figure 15.2a illustrates a wormhole connecting the Kwik-E-Mart and the Springfield Nuclear Power Plant, but the drawing is misleading because the wormhole appears to stretch across Springfield airspace. More accurately, the wormhole should be thought of as a new region of space that interfaces with ordinary, familiar space only at its ends—its mouths. If while walking along the streets of Springfield, you scoured the skyline in search of the wormhole, you'd see nothing. The only way to see it would be to hop on over to the Kwik-E-Mart, where you would find an opening in ordinary space—one wormhole mouth. Looking